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Linear Integral Analysis of Bar Rough Rolling by Strain Rate Vector

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Abstract: A new linear integral method for bar hot rolling on roughing train was obtained. First, for plastic deformation energy rate, equivalent strain rate about Kobayashi's three-dimensional velocity field was expressed by two-dimensional strain rate vector; then, the two-dimensional strain rate vector was inverted into inner product and was integrated term by term. During those processes, boundary equation and mean value theorem were introduced; for friction and shear energy dissipation rate, definite integral was applied to the solution process. Sequentially, the total upper bound power was minimized, and the analytical expressions of rolling torque, separating force, and stress state factor were obtained. The calculated results by these expressions were compared with those of experimental values. The results show that the new linear integral method is available for bar rough rolling analysis and the calculated results by this method are a little higher than those of experimental ones. However, the maximum error between them is less than 10%.

Key words: bar rough rolling; linear integration; strain rate vector; roll separating force

Hot rolling, using shaped or grooved rolls, is one of the methods of producing bars and rods. Introduced from America, Morgan high-speed no-twist wire and rod mill is made up of 25 stands. Rough rolling unit refers to the first five stands of wire and rod continuous mill. In shape rolling, the deformation behavior of the billet is more complex than that of hot slabs and plates. Simplified one-dimensional models, derived for the calculation of rolling force and torque and mostly based on empirical modifications of the flat rolling theories, have been reported[1-3]. Since the quadratic equations of interface and free-edge surface (unconfined surface) of the deformation zone are complex, studies on analytical solution of total energy rate for three-dimensional rolling are very difficult[4]. Although a comprehensive analysis was given by Kobayashi in 1975[5] and afterward several finite element methods were rapidly developed[6-11], there is still no analytical solution without simplifying above equations for the rolling. Therefore, a new linear integral method is applied to Kobayashi's velocity fields, to obtain analytical solutions of total energy rate, roll separating force and torque from Stand 1 to Stand 5 of the roughing train, and the calculated results by the expressions of analytical solutions are compared with those of experimental ones[12].

1 Velocity Field

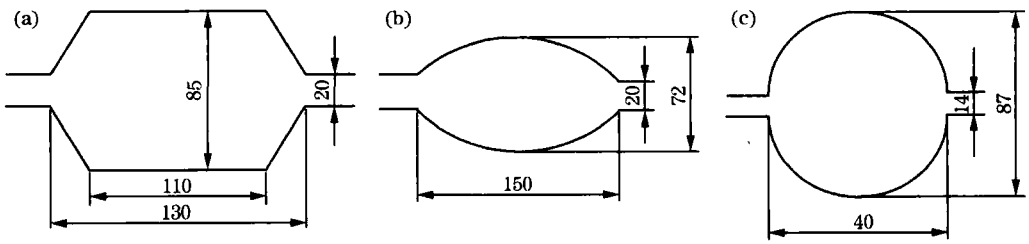
The billet with dimensions of 115 mm × 115 mm is rolled through a pair of cylindrical rolls with radius of r_0, which enables the thickness of the billet to be reduced from $2h_0$ to $2h_1$ and the width increased from $2b_0$ ($=2h_0$) to $2b_1$. The shapes and sizes of No. 1 to No. 3 square-oval-round grooves are shown in Fig. 1.

The projected length of the contact arc between the roller and the billet is given by l, as shown in Fig. 2 (a). Coordinate axes are taken with the origin at the midpoint of the entry cross-section of deformation zone. b_x and h_x (h_a) are half of spread and thickness when horizontal distance from the origin is x. Owing to the symmetry of deformation zone, only a quarter is considered, as shown in Fig. 2 (b).

For the three-dimensional rolling, Hill and Kobayashi gave the following velocity field[5,13-14]:

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(a) No. 1 pass; (b) No. 2 pass; (c) No. 3 pass.

Fig. 1 The shapes and sizes of grooves for No. 1 to No. 3 pass

$$v_x=\frac{U}{h_x b_x},\quad v_y=v_x\frac{b'_x y}{b_x},\quad v_z=v_x\frac{h'_x}{h_x}z \tag{1}$$

where

$$U=h_x b_x v_x=h_0 b_0 v_0=h_1 b_1 v_1=v_r\cos\alpha_n h_n b_n$$

$$h_x=r_0+h_1-\sqrt{r_0^2-(l-x)^2}$$

$$b_x=b_0+\frac{\Delta b x}{l}$$

$$\Delta b=b_1-b_0$$

v_x, v_y, v_z are the velocity components in x, y, z directions; U is the flow volume per second; v_r is the tangential velocity of the roll at groove bottom; α_n, h_n, b_n are angle, thickness, and width at neutral plane in deforming zone.

According to Cauchy equation, the strain rate components from Eqn. (1) are:

$$\begin{cases}\dot\varepsilon_x=-v_x\left(\dfrac{b'_x}{b_x}+\dfrac{h'_x}{h_x}\right),\ \dot\varepsilon_y=\dfrac{v_x}{b_x}b'_x,\ \dot\varepsilon_z=\dfrac{v_x}{h_x}h'_x,\\ \dot\varepsilon_{xy}=\dfrac{v_x}{2}\left[\dfrac{b''_x}{b_x}-\dfrac{h'_x b'_x}{h_x b_x}-2\left(\dfrac{b'_x}{b_x}\right)^2\right]y,\\ \dot\varepsilon_{zx}=\dfrac{v_x}{2}\left[\dfrac{h''_x}{h_x}-\dfrac{h'_x b'_x}{h_x b_x}-2\left(\dfrac{h'_x}{h_x}\right)^2\right]z\end{cases} \tag{2}$$

It can be validated that Eqn. (1) and Eqn. (2) satisfy the velocity boundary condition, incompressibility, and steady-state rolling condition.

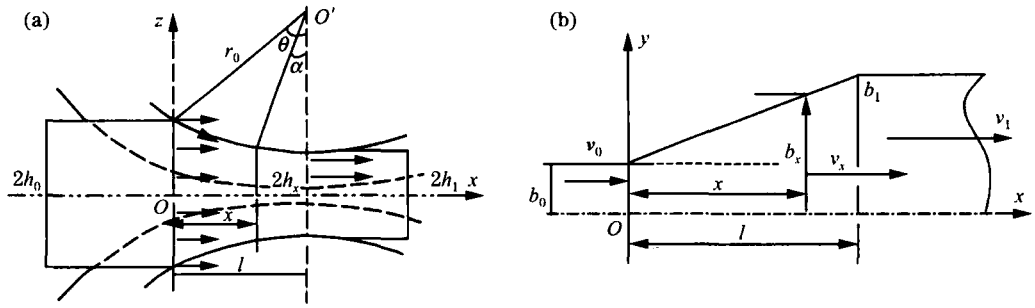


Fig. 2 Deformation zone of the first stand (a) and half of spread in deforming zone (b)

2 Total Energy Rate

By using the first variation principle[14] of rigid-plastic materials, the total energy rate function Φ should be minimized.

$$\Phi=N_d+N_f+N_s \tag{3}$$

where, N_d is the plastic deformation energy rate; N_f is the friction energy rate; and N_s is the shear energy dissipation rate.

2.1 Integration of plastic deformation energy rate

Substituting $\bar\sigma=R_e$ and $\dot{\bar\varepsilon}=\left[\frac{2}{3}(\dot\varepsilon_x^2+\dot\varepsilon_y^2+\dot\varepsilon_z^2+2\dot\varepsilon_{xy}^2+2\dot\varepsilon_{xz}^2)\right]^{1/2}$ into the following equation and integrating the plastic deformation energy rate[14], then

$$N_d=\int_V\bar\sigma\,\dot{\bar\varepsilon}\,dV=4\sqrt{\frac{2}{3}}R_e\int_V\sqrt{\dot\varepsilon_{ij}\dot\varepsilon_{ij}}\,dV=4\sqrt{\frac{2}{3}}R_e\cdot\int_0^{h_x}\int_0^{b_x}\int_0^{l}(v_x^2g^2+v_x^2H^2y^2+v_x^2I^2z^2)^{1/2}dxdydz=$$

$$4\sqrt{\frac{2}{3}}R_e\int_0^{h_x}\int_0^{b_x}\int_0^{l}|\dot{\boldsymbol\varepsilon}|\,|\dot{\boldsymbol\varepsilon}_0|\cos(\dot{\boldsymbol\varepsilon},\dot{\boldsymbol\varepsilon}_0)dxdydz=$$

$$4\sqrt{\frac{2}{3}}R_e\int_0^{h_x}\int_0^{b_x}\int_0^{l}\dot{\boldsymbol\varepsilon}\cdot\dot{\boldsymbol\varepsilon}_0\,dxdydz \tag{4}$$

where

$$\dot{\boldsymbol\varepsilon}=gv_x\boldsymbol i+Hyv_x\boldsymbol j+Izv_x\boldsymbol k$$

$$\dot{\boldsymbol\varepsilon}_0=e_1\boldsymbol i+e_2\boldsymbol j+e_3\boldsymbol k$$

$$g=\sqrt{2\left[\left|\frac{b'_x}{b_x}\right|^2+\left|\frac{h'_x}{h_x}\right|^2+\left|\frac{h'_x}{h_x}\right|\left|\frac{b'_x}{b_x}\right|\right]}$$

$$H=\frac{1}{\sqrt{2}}\left[\frac{b''_x}{b_x}-\frac{h'_x b'_x}{h_x b_x}-2\left(\frac{b'_x}{b_x}\right)^2\right]$$

$$I=\frac{1}{\sqrt{2}}\left[\frac{h''_x}{h_x}-\frac{h'_x b'_x}{h_x b_x}-2\left|\frac{h'_x}{h_x}\right|^2\right] \tag{5}$$

$\dot{\boldsymbol{\varepsilon}}_0$ is the unit vector of $\dot{\boldsymbol{\varepsilon}}$.

Take cosine of projection about the unit vector of $\dot{\boldsymbol{\varepsilon}}$ at axis x, y, z directions as e_1, e_2, e_3, then the following equations can be obtained:

$$e_1=\frac{\mathrm{d}x}{\sqrt{\mathrm{d}x^2+\mathrm{d}y^2+\mathrm{d}z^2}},\quad e_2=\frac{\mathrm{d}y}{\sqrt{\mathrm{d}x^2+\mathrm{d}y^2+\mathrm{d}z^2}},$$
$$e_3=\frac{\mathrm{d}z}{\sqrt{\mathrm{d}x^2+\mathrm{d}y^2+\mathrm{d}z^2}} \tag{6}$$

Substituting Eqn. (5) and Eqn. (6) into Eqn. (4), then

$$N_{\mathrm{d}}=4\sqrt{\frac{2}{3}}R_{\mathrm{e}}\int_0^{h_x}\int_0^{b_x}\int_0^{l}\left[\frac{gv_x\mathrm{d}x\mathrm{d}y\mathrm{d}z}{\sqrt{1+\left|\frac{\mathrm{d}y}{\mathrm{d}x}\right|^2+\left|\frac{\mathrm{d}z}{\mathrm{d}x}\right|^2}}+\right.$$
$$\left.\frac{Hyv_x\mathrm{d}x\mathrm{d}y\mathrm{d}z}{\sqrt{1+\left|\frac{\mathrm{d}x}{\mathrm{d}y}\right|^2+\left|\frac{\mathrm{d}z}{\mathrm{d}y}\right|^2}}+\frac{Izv_x\mathrm{d}x\mathrm{d}y\mathrm{d}z}{\sqrt{1+\left|\frac{\mathrm{d}x}{\mathrm{d}z}\right|^2+\left|\frac{\mathrm{d}y}{\mathrm{d}z}\right|^2}}\right]=$$
$$4\sqrt{\frac{2}{3}}R_{\mathrm{e}}(I_1+I_2+I_3) \tag{7}$$

Integrating Eqn. (7) by mean value theorem term by term, I_1, I_2, and I_3 can be obtained as follows[15]:

$$I_1=\sqrt{2}lU\left[\frac{\left|\frac{\Delta b}{b_m}\right|^2+\left|\frac{\Delta h}{h_m}\right|^2-\frac{\Delta b\Delta h}{b_m h_m}}{l^2+\Delta b^2+\Delta h^2}\right]^{1/2}=\sqrt{2}lUf_1$$
$$I_2=\frac{lU}{2\sqrt{2}}\frac{\frac{(\Delta b)^2\Delta h}{l^2h_m}-\frac{2(\Delta b)^3}{l^2b_m}}{\sqrt{l^2+\Delta b^2}}=\frac{lU}{2\sqrt{2}}f_2$$
$$I_3=\frac{-\frac{lU}{2\sqrt{2}}\left[\frac{\Delta h}{R_0-\Delta h}+\frac{\Delta b\Delta h^2}{l^2b_m}-2\frac{\Delta h^3}{l^2h_m}\right]}{\sqrt{l^2+\Delta h^2}}=\frac{lUf_3}{2\sqrt{2}}$$

Substituting I_1, I_2, and I_3 into Eqn. (7) gives

$$N_d=4\sqrt{\frac{2}{3}}R_{\mathrm{e}}\left[l\sqrt{2}Uf_1+\frac{Ul}{2\sqrt{2}}f_2+\frac{Ul}{2\sqrt{2}}f_3\right] \tag{8}$$

2.2 Surface integration of friction energy rate

The friction energy rate on the interface between the roller and the work piece deduced by Kobayashi in 1975[5,14] is:

$$N_{\mathrm{f}}=\int_0^{l}\int_0^{b_x}\tau_{\mathrm{f}}|\Delta v_{\mathrm{f}}|\mathrm{d}s \tag{9}$$

where

$\tau_{\mathrm{f}}=mk$

$$|\Delta v_{\mathrm{f}}|=U\sqrt{\left|\frac{b'_x y}{b_x^2h_x}\right|^2+\left[\frac{v_r}{U}-\frac{\sqrt{1+(h'_x)^2}}{b_xh_x}\right]^2}$$

τ_{f} is the friction shear stress. Noticing that along the tangential direction of the roll surface, τ_{f} is the co-line vector with the tangential velocity discontinuity Δv_{f}, and the roll surface equation is:

$$z=h_x=r_0+h_1-[r_0^2-(l-x)^2]^{1/2}$$

Integrating Eqn. (9)[15-16] leads to

$$N_{\mathrm{f1}}=4mk\left\{\frac{\Delta br_0^2v_r}{l}(1+\cos\theta-2\cos\alpha_{\mathrm{n}})+v_rr_0b_1(\theta-\right.$$
$$\left.2\alpha_{\mathrm{n}})+\frac{Ur_0}{h_m}\ln\frac{\tan^2\left|\frac{\pi}{4}+\frac{\alpha_{\mathrm{n}}}{2}\right|}{\tan\left|\frac{\pi}{4}+\frac{\theta}{2}\right|}\right\} \tag{10}$$

where N_{f1} is the friction energy rate on the contact surface between the roller groove bottom and the workpiece.

2.3 Integral depending on shear energy dissipation rate

The tangential velocity discontinuity at exit cross-section of deforming zone is

$$v_z|_{x=l}=\Delta v_z|_{x=l}=0,\ v_y|_{x=l}=\Delta v_y|_{x=l}=\frac{\Delta b}{l}v_x|_{x=l}=\frac{\Delta bv_1}{l}\frac{y}{b_1}$$

And shear energy dissipation rate on this section is

$$N_{\mathrm{s1}}=4k\int_0^{b_1}\int_0^{h_1}\sqrt{\Delta v_y^2+\Delta v_z^2}\mathrm{d}z\mathrm{d}y=\frac{2kU\Delta b}{l} \tag{11}$$

where N_{s1} is the shear energy dissipation rate on the exit cross-section.

On the entry section, the tangential velocity discontinuity from Eqn. (1) is

$$v_y|_{x=0}=\frac{v_0\Delta b}{lb_0}y,\ v_z|_{x=0}=-\frac{v_0\tan\theta}{h_0}z$$
$$\bar{v}_y|_{x=0}=\frac{1}{b_0}\int_0^{b_0}\frac{v_0\Delta b}{lb_0}y\mathrm{d}y=\frac{v_0\Delta b}{2l}$$
$$\bar{v}_z|_{x=0}=\frac{1}{h_0}\int_0^{h_0}-\frac{v_0\tan\theta}{h_0}z\mathrm{d}z=-\frac{v_0\tan\theta}{2}$$
$$\frac{\bar{v}_z}{\bar{v}_y}|_{x=0}=-\frac{l\tan\theta}{\Delta b}$$

Over the whole domain at entry section in Fig. 3, mean value theorem was introduced to integrate Eqn. (11) by definite integral to get an analytical result,

$$N_{\mathrm{s0}}=4k\int_0^{b_0}\int_0^{h_0}[\sqrt{v_y^2+v_z^2}]_{x=0}\mathrm{d}z\mathrm{d}y=$$

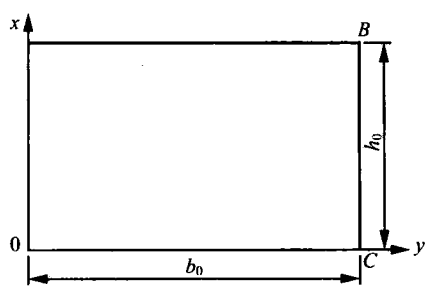


Fig. 3 Integral domain of entry section

$$\frac{2kU\Delta b}{l}\sqrt{1+\frac{l^2\tan^2\theta}{\Delta b^2}} \tag{12}$$

where N_{s0} is the shear energy dissipation rate on the entry cross-section.

2.4 Friction losses at sidewall of roller

The tangential velocity discontinuity between sidewall of roller is

$$\Delta v_{f2}=v_x-v_r\cos\alpha \tag{13}$$

where

$v_r=\omega r_0$,

$\Delta r=r_h-r_0$

$\bar{v}=(v_1+v_0)/2$

$\bar{r}=(r_h+r_0)/2$

ω is the angular velocity of the roll.

The friction losses on the surface is,

$$N_{f2}=4mk\int_{\alpha_n}^{\theta}\int_{r_0}^{r_h}\left|\frac{v_x}{\cos\alpha}-v_r\right|r\,dr\,d\alpha+4mk\int_{\alpha_n}^{0}\int_{r_0}^{r_h}\left|\frac{v_x}{\cos\alpha}-v_r\right|r\,dr\,d\alpha \tag{14}$$

where N_{f2} is the friction energy rate on the contact surface between the roller groove sidewall and the workpiece.

Assuming the velocity in the zone of slippage on the entry and exit sides as $v_x\approx\frac{v_0+v_r}{2}$ and $v_x\approx\frac{v_1+v_r}{2}$, Eqn. (14) becomes

$$N_{f2}=4mk\,\bar{r}\Delta r[(v_0+v_r)\ln\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)-2(\bar{v}+v_r)\ln\left(\frac{\pi}{4}+\frac{\alpha_n}{2}\right)-2v_r(\theta-2\alpha_n)] \tag{15}$$

2.5 Total energy rate and its minimization

Substituting Eqn. (8), Eqn. (10), Eqn. (11), Eqn. (12), and Eqn. (15) into Eqn. (3), and considering $N_s=N_{s0}+N_{s1}$ and $N_f=N_{f1}+N_{f2}$, then

$$\Phi=N_d+N_{s0}+N_{s1}+N_{f1}+N_{f2}=4\sqrt{\frac{2}{3}}R_e[l\sqrt{2}Uf_1+\frac{Ul}{2\sqrt{2}}f_2+\frac{Ul}{2\sqrt{2}}f_3]+\frac{2kU\Delta b}{l}+\frac{2kU\Delta b}{l}\sqrt{1+\frac{l^2\tan^2\theta}{\Delta b^2}}+4mk\{\frac{\Delta br_0^2v_r}{l}\cdot(1+\cos\theta-2\cos\alpha_n)+v_rr_0b_1(\theta-2\alpha_n)+\frac{Ur_0}{h_m}\ln\frac{\tan^2\left(\frac{\pi}{4}+\frac{\alpha_n}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)}\}+4mk\,\bar{r}\Delta r[(v_0+v_r)\cdot\ln\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)-2(\bar{v}+v_r)\ln\tan\left(\frac{\pi}{4}+\frac{\alpha_n}{2}\right)-2v_r(\theta-2\alpha_n)] \tag{16}$$

Noticing $h_n=r_0+h_1-r_0\cos\alpha_n$, $b_n=b_0+\frac{\Delta bx_n}{l}$, $x_n=l-r\sin\alpha_n$,

$$U=(v_rr_0b_1+v_rh_1b_1)\cos\alpha_n-b_1r_0v_r\cos^2\alpha_n-\left(\frac{v_r\Delta br_0^2}{l}+\frac{v_rh_1\Delta br_0}{l}\right)\cos\alpha_n\sin\alpha_n+\frac{v_r\Delta br_0^2}{l}(\sin\alpha_n-\sin^3\alpha_n)$$

Taking $a=\frac{v_rr_0^2\Delta b}{l}$, $b=1+\frac{h_1}{r_0}$, $c=\frac{b_1l}{r_0\Delta b}$ and differentiating U with respect to neutral angle α_n, then

$$\frac{dU}{d\alpha_n}=-abc\sin\alpha_n+ac\sin2\alpha_n-ab\cos2\alpha_n-2a\cos\alpha_n+3a\cos^3\alpha_n \tag{17}$$

Differentiating the total energy Φ of Eqn. (16) with respect to neutral angle α_n and setting it to be zero, the following equation can be obtained:

$$\frac{d\Phi}{d\alpha_n}=\frac{\partial N_d}{\partial\alpha_n}+\frac{\partial N_{s0}}{\partial\alpha_n}+\frac{\partial N_{s1}}{\partial\alpha_n}+\frac{\partial N_{f1}}{\partial\alpha_n}+\frac{\partial N_{f2}}{\partial\alpha_n}=0 \tag{18}$$

Differentiating Eqn. (18) term by term and taking $dU/d\alpha_n=N$, the analytical expression of m is

$$m=\left[H\left(4lf_1+\frac{lf_2}{2}+\frac{lf_3}{2}\right)+\frac{H\Delta bf_4}{2l}\right]\div\{2v_rr_0b_1-\bar{r}\Delta r\left[2v_r-\frac{(\bar{v}+v_r)}{\cos\alpha_n}\right]-\frac{2\Delta br_0^2v_r\sin\alpha_n}{l}-\frac{Hr_0}{h_m}\ln\frac{\tan^2\left(\frac{\pi}{4}+\frac{\alpha_n}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)}-\frac{2Ur_0}{h_m\cos\alpha_n}\} \tag{19}$$

Substituting α_n determined by Eqn. (19) into Eqn. (16), the minimum total energy rate Φ_{min} is obtained.

Noticing that the average width of deforming zone is $\bar{B}=2\bar{b}_x$, and the contact area is $F=2\bar{b}_xl$, then the power required can be written as

$$J=4\bar{p}\,\bar{b}_xl\,\frac{1}{l}\int_0^l v_{rz}\,dx=\frac{2\bar{p}\,\bar{b}_xl^2v_r}{r_0}=\Phi_{min} \tag{20}$$

where, $v_{rz}=-v_r\sin\alpha$, $\bar{p}=\frac{r_0}{2\bar{b}_xl^2v_r}\Phi_{min}$, and $\bar{p}$ is average unit pressure.

Through Φ_{min}, the corresponding expressions of rolling torque (M), roll separating force (P), and stress effective factor (n_σ) are given by

$$P=\bar{p}\cdot F,\ M=\frac{r_0}{2v_r}\Phi_{min},\ n_\sigma=\frac{\bar{p}}{R_e}=\frac{r_0\Phi_{min}}{2\bar{b}_xl^2v_rR_e} \tag{21}$$

Or using Tselicov formula[17-18],

$$\bar{p}=\frac{\Phi_{min}}{v_r8\chi b_m\Delta h},\ M=\frac{r_0\Phi_{min}}{2v_r},\ n_\sigma=\frac{\bar{p}}{R_e} \tag{22}$$

where χ is an arm factor, and for hot rolling, $\chi=0.3-0.6$.

3 Discussion of Results

Morgan data from Stand 1 to Stand 5[12] are shown in Table 1. S80 high-quality carbon steel billet with dimensions of 115 mm × 115 mm was rolled through first strand of the roughing train at 1050 ℃. The deformation resistance[19] is given by

$$R_e = 135.4\exp(3.751-2.974T)\cdot\left|\frac{\dot{\varepsilon}}{10}\right|^{0.034T+0.1089}\cdot\left[1.312\left|\frac{\varepsilon}{0.4}\right|^{0.2781}-0.78\varepsilon\right] \quad (23)$$

where $T=\frac{t+273}{1000}$; and t is the deformation temperature.

With the equations above, programming by C language and using golden section search method, the rolling torque and roll separating force are minimized with $m=1.0$. The results are shown in Table 2.

It shows that the rolling force and torque assignment among the first five stands are nonuniform, owing to the uneven of reductions in passes. Therefore, it is necessary to increase the reduction of the second pass to balance the loading of each stand.

1) Comparison between the calculated values and the measured ones

According to Table 2, the comparisons of the calculated roll separating force and rolling torque with the measured ones[12] are shown in Fig. 4. Fig. 4 (a) indicates that the calculated rolling forces are slightly higher than the measured, for this is an upper bound method. But the errors are less than 10%. And Fig. 4 (b) shows that the calculated rolling torques approach to the measured values. Thus, it can be concluded that this new linear integration is available for bar rough rolling analysis.

2) Neutral angle and stress effective factor versus l/h for various values of m

As shown in Fig. 5, neutral angle increases with the increase in friction when $l/h=0.8$; when the

Table 1 Morgan data of rough stands

Stand	Shape	Roll gap/ mm	Depth/ mm	Height/ mm	Roll diameter/ mm	Motor speed/ (r · min^{-1})	Gear ratio/ %	Exit speed/ (m · s^{-1})
1	Square	20	32.5	85	460	283.65	36.14	0.1663
2	Oval	20	26	72	483	306.27	34.071	0.2132
3	Round	14	36.5	87	457	492.33	34.679	0.2978
4	Oval	14	17.5	49	480	335.92	18.774	0.4322
5	Round	10	26	62	479	326.32	12.772	0.5865

Table 2 Calculated results of roughing train

Stand	1RM	2RM	3RM	4RM	5RM
Reduction/%	26.1	15.3	40.7	43.7	41.7
n_σ	1.2134	0.9783	1.1869	1.3912	1.2832
l/h	0.80	0.87	1.02	1.50	1.28
Rolling force/kN	677.3	419.4	456.1	677.0	502.1
Rolling torque/(kN · m)	71.2	39.0	62.0	41.1	51.6

Note: $h=(h_1+h_0)/2$.

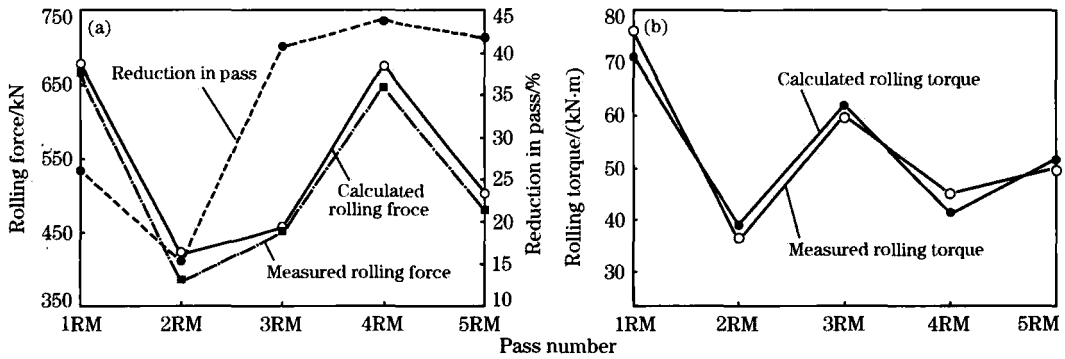


Fig. 4 Relationship among the measured and calculated roll separating force and reduction (a) and rolling torque (b) and roll numbers

value of m is less than 0.5, the relation between α_n and m is nearly linear. The value of n_σ decreases at first and then increases with increasing the value of l/h and m, as shown in Fig. 6. When the value of l/h is close to 0.9, n_σ reaches the minimum value.

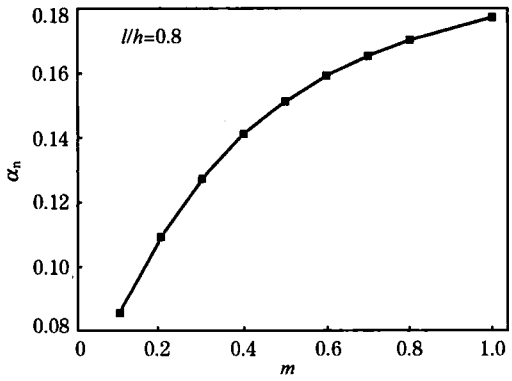


Fig. 5 The relationship between neutral angle α_n and friction factor m

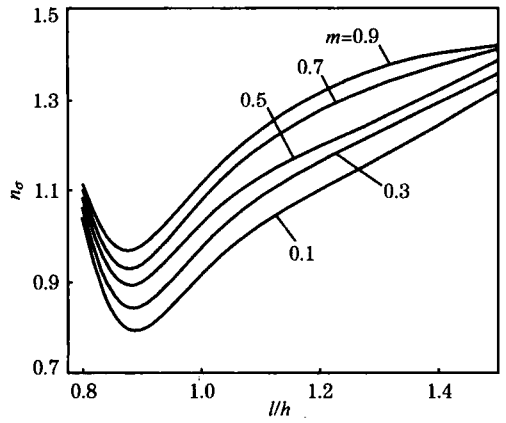


Fig. 6 Relationship between stress effective factor n_σ and l/h for various values of m

4 Conclusions

1) A new linear integral method is proposed for bar rolling analysis. The calculation during rough rolling shows that the calculated roll separating forces are a little higher than the measured ones, and the rolling torque values approach the measured ones, but the maximum error is less than 10%.

2) The reduction of the first five stands is non-uniform, which results in the uneven rolling force and torque among these stands. Therefore, it is necessary to increase the reduction of the second pass and to balance the loading of each stand.

3) The solution above is available for bar rolling in the roughing train, and after stand 6, the error percentage will be significant.

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